

OPTIONS: CONTRACTS, PAYOFFS, AND THE GREEKS

FROM THE RIGHT TO TRADE, TO THE RISK YOU HEDGE

Financial Econometrics

WHAT THIS LECTURE BUILDS

An option is a contract whose value depends, above all, on **volatility** — the quantity the rest of the course has spent its time modelling. Here we meet the instrument itself, before pricing it.

- What an option *is*, and the four building-block positions.
- How options *vary*, and what an option is *worth* — at expiry and before.
- How to read an **option chain**.
- The **Greeks** — the sensitivities that are the daily language of risk — one at a time.
- **Put–call parity**, the no-arbitrage tie between calls and puts.

The thread to keep in view

An option is, at its core, a **bet on volatility**. Watch for where σ enters.

OUTLINE

- ➊ What is an option?
- ➋ Variations of options
- ➌ Payoffs and profit at expiry
- ➍ What an option is worth before expiry
- ➎ Reading an option chain
- ➏ The Greeks, line by line
- ➐ Put–call parity
- ➑ A first look at continuous-time models
- ➒ Synthesis

WHAT IS AN OPTION?

Definition

An **option** gives its holder the **right, but not the obligation**, to trade an underlying asset at a fixed **strike price** K by a fixed **expiry** T , in exchange for a price paid up front — the **premium**.

Two kinds:

- A **call** gives the right to **buy** at K (you want the price to *rise*).
- A **put** gives the right to **sell** at K (you want the price to *fall*).

The key word is *right*: the holder exercises only when it pays, and walks away otherwise. That one-sidedness is what the premium buys.

TWO SIDES: THE BUYER (LONG)

Every option is a contract between two parties with *opposite* rights and obligations. Start with the party who **pays**.

Buyer — long

- **Pays** the premium up front.
- Owns the **right** to exercise — and does so only when it pays, walking away otherwise.
- **Max loss** = the premium; **upside** large (unbounded for a call).

The buyer has bought **asymmetry**: a known, capped cost in exchange for a large potential gain. That asymmetry is exactly what the premium pays for.

TWO SIDES: THE WRITER (SHORT)

The other side of every contract is the party who **receives** the premium.

Writer — short

- **Receives** the premium up front.
- Carries the **obligation** to deliver *if assigned*.
- **Max gain** = the premium; **downside** large — so the writer must post **margin** (collateral).

Exercise vs. assignment: the *holder* chooses to exercise; a writer is then *assigned* and must deliver (American: any time; European: only at expiry).

BUYER AND WRITER: A ZERO-SUM CONTRACT

The key idea

Before fees, the option is **zero-sum**: the buyer's payoff is exactly the writer's loss. The premium is the *price of the asymmetry* — what the buyer pays for limited loss and open upside, and the writer's compensation for taking the other side.

This tension — a capped-risk buyer against a premium-earning writer — runs through everything that follows: the payoff diagrams, the Greeks, and every strategy built from them.

THE FOUR BUILDING BLOCKS

Two contract types $\times$ two sides = the four positions every strategy is built from:

Position	Market view	Premium	Max gain	Max loss
Long call	bullish	pay	unbounded	premium
Long put	bearish	pay	large (K less prem.)	premium
Short call	bearish / neutral	receive	premium	unbounded
Short put	bullish / neutral	receive	premium	large (K less prem.)

- **Buyers** (long) pay to get large upside and capped loss.
- **Writers** (short) collect the premium but carry the large, open-ended risk.

Spreads, straddles, collars — every strategy is a **combination** of these four. We will see their shapes next.

HOW OPTIONS VARY

Those four positions are the atoms; real contracts vary along several dimensions.

Dimension	Variants
Type	call (right to buy) vs put (right to sell)
Exercise style	European (only at expiry) vs American (any time)
Side	long (holder) vs short (writer)
Moneyness	ITM , ATM , OTM
Maturity	short-dated, or long-dated LEAPS (a year or more)
Payoff	vanilla (standard) vs exotic (barrier, Asian, lookback, ...)

Two of these matter most for us: **exercise style** and **moneyness**.

EXERCISE STYLE AND MONEYNES

Exercise style.

- **American** — exercisable early, so (weakly) worth more and harder to price. Most single-stock options are American.
- **European** — exercisable only at expiry. Many index options (including S&P 500 options) are European. *Black–Scholes prices European options.*

Moneyness — where the strike sits relative to the spot S :

	Call	Put
In-the-money (ITM)	$S > K$	$S < K$
At-the-money (ATM)	$S \approx K$	$S \approx K$
Out-of-the-money (OTM)	$S < K$	$S > K$

Moneyness drives almost everything about how an option behaves — as the Greeks will show.

PAYOFFS AT EXPIRY

Start at the one moment when an option's value is beyond dispute — **expiry**, where only the payoff remains:

$$\text{call payoff} = \max(S_T - K, 0), \quad \text{put payoff} = \max(K - S_T, 0).$$

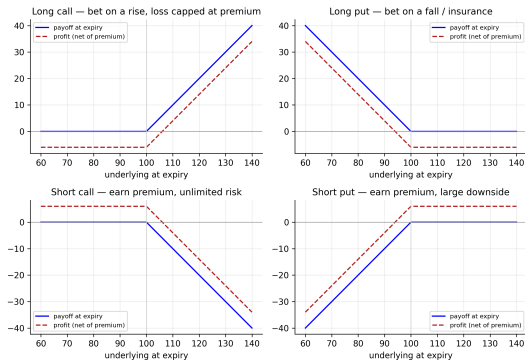
- The holder's **profit** subtracts the premium paid.
- The writer's profit is the **mirror image**: premium received, minus any payoff owed.

Why start here

The payoff is the contract's DNA. Everything before expiry — price, Greeks, hedging — is an expectation *about* this payoff.

THE FOUR POSITIONS, DRAWN

The four basic option positions (strike $K=100$)



Blue = payoff at expiry; red = profit after the premium (strike $K = 100$).

READING THE FOUR RISK PROFILES

- **Long call** — bullish; loss capped at the premium, *unbounded* upside.
- **Long put** — profits as the price falls; the classic way to **insure** a holding.
- **Short call** — collects the premium, but faces *unlimited* loss if the price soars.
- **Short put** — collects the premium, but is exposed to a large drop.

The asymmetry

Buyers have limited loss and large upside; writers have limited gain (the premium) and large risk. These asymmetries are exactly what the **Greeks** quantify.

INTRINSIC VALUE + TIME VALUE

Before expiry, an option's price splits in two:

$$\underbrace{\text{option price}}_{\text{premium}} = \underbrace{\text{intrinsic value}}_{\text{payoff if exercised now}} + \underbrace{\text{time value}}_{\text{extra, for the chance of doing better}} .$$

- **Intrinsic value** — worth if exercised immediately, e.g. $\max(S - K, 0)$ for a call.
- **Time value** — the premium on top, for the chance of finishing *deeper* in the money. Largest for **ATM** options; decays to zero as expiry approaches.

SIX INPUTS — AND THE ONE THAT MATTERS MOST

The total price is set by six inputs:

S (spot), K (strike), T (time), r (rate), dividends, σ (volatility).

Volatility raises *both* calls and puts

Higher σ widens the range of outcomes. That makes the valuable tail more likely *without* hurting the capped side — so both calls and puts become more expensive.

This is why the entire volatility-modelling half of the course feeds straight into option pricing: five of the six inputs are observable; σ is the one you must *forecast*.

WHAT AN OPTION CHAIN IS

Those six inputs set the price of a *single* option. In practice a trader watches dozens at once — every strike for a given expiry, side by side. That table is the **option chain**.

For one expiry, it lists every strike alongside:

- call and put prices (**bid**, **ask**, **last**);
- trading **volume** and **open interest** (contracts outstanding);
- the **implied volatility** backed out of each market price.

AN OPTION CHAIN, READ DOWN THE STRIKES

S&P 500, spot = 7483, $\sigma = 17.4\%$, 3-month, $r = 4\%$. Prices in index points.

Strike	Money.	Call	Put	$\Delta(\text{call})$	Γ/pt	Vega/ $1\%\sigma$	Θ/day
6735	ITM	841.5	26.2	0.92	0.00024	5.8	-1.21
7109	ITM	534.0	89.0	0.77	0.00046	11.3	-1.65
7483	ATM	296.8	222.1	0.56	0.00061	14.7	-1.83
7857	OTM	142.2	437.8	0.34	0.00057	13.8	-1.58
8232	OTM	58.2	725.1	0.17	0.00040	9.6	-1.05

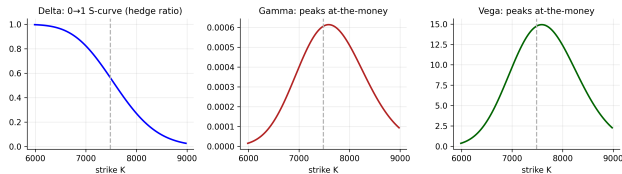
- Deep **ITM call** ($K = 6735$): $\approx$ its intrinsic value + a little; behaves like the stock ($\Delta = 0.92$).
- **OTM call** ($K = 8232$): a cheap lottery ticket ($\Delta = 0.17$).
- **ATM option**: the *most* time value and the *largest* Greeks — which brings us to the Greeks.

THE GREEKS

Definition

The **Greeks** are an option's sensitivities — how its price moves with the underlying (Δ), with volatility (ν), with time (Θ), and so on. Running an options book *is* managing the Greeks: you hedge each sensitivity you do not want.

S&P 500 call Greeks across strikes (vertical line = spot)



Delta runs $0 \rightarrow 1$ in an S-curve; **gamma** and **vega** are bell-shaped, peaking **at-the-money**. We take them one at a time, using the S&P ATM call for concrete numbers.

DELTA (Δ) — SENSITIVITY TO THE UNDERLYING

$$\Delta = \frac{\partial(\text{price})}{\partial S}, \quad \Delta_{\text{call}} = N(d_1) \in [0, 1], \quad \Delta_{\text{put}} = N(d_1) - 1.$$

- **The hedge ratio:** shares held to move one-for-one with the option. The S&P ATM call has $\Delta = 0.56$ — sell it, hedge by buying 0.56 index units per option (**delta hedging**).
- Doubles as a rough **probability of finishing ITM**: deep-ITM $\Delta \rightarrow 1$ (it *is* the stock), deep-OTM $\Delta \rightarrow 0$.

Why it matters: delta is a position's directional exposure; a “delta-neutral” book has no first-order bet on where the market goes.

GAMMA (Γ) — HOW FAST DELTA CHANGES

$$\Gamma = \frac{\partial \Delta}{\partial S} = \frac{\partial^2(\text{price})}{\partial S^2} = \frac{\phi(d_1)}{S \sigma \sqrt{T}}.$$

- The **curvature**: the rate at which delta itself moves. **Peaks at-the-money** and near expiry (S&P ATM call $\Gamma = 0.00061$).
- High gamma $\Rightarrow$ constant **re-hedging**: every move changes the delta.

Why it matters: near expiry an ATM option's delta can swing violently — gamma is the source of the trickiest risk in a book. Long-gamma profits from big moves; short-gamma (option sellers) is hurt by them.

VEGA (ν) — SENSITIVITY TO VOLATILITY

$$\nu = \frac{\partial(\text{price})}{\partial\sigma} = S \phi(d_1) \sqrt{T}.$$

- Exposure to **volatility itself**: the S&P ATM call gains ≈ 14.7 points per 1% of volatility. Like gamma, it **peaks at-the-money**.

The bridge to the whole course

An option is a **bet on volatility**, and vega is that exposure. Because σ is the one *unobservable* input, **forecasting volatility is managing vega** — every ARCH / GARCH / *t*-GARCH model is, in the options world, a tool for pricing and hedging vega.

THETA (Θ) — TIME DECAY

$$\Theta = \frac{\partial(\text{price})}{\partial t} < 0 \quad (\text{for a long option}).$$

- The **bleed of time value**: with each day there is less chance left to move in your favour, so a long option *loses* value even if nothing else changes (S&P ATM call ≈ -1.8 points/day).
- **Buyers pay theta** (time works against them); **sellers earn it**.

The **theta-vega trade-off** — you cannot collect time decay without being short volatility — is the central tension of most option strategies.

RHO (ρ) — SENSITIVITY TO INTEREST RATES

$$\rho = \frac{\partial(\text{price})}{\partial r}.$$

- The S&P ATM call gains ≈ 9.8 points per 1% rise in rates; the put loses about the same.
- Usually the **smallest** Greek for short-dated options and often ignored...
- ...but it grows with maturity — it matters for **long-dated** options (LEAPS) and in rate-sensitive markets.

THE FIVE GREEKS AT A GLANCE

Greek	Sensitivity to	ATM call	Meaning
Δ Delta	underlying S	0.56	hedge ratio; directional exposure
Γ Gamma	delta itself	0.0006/pt	re-hedging cost; largest ATM
ν Vega	volatility σ	14.7/vol-pt	the volatility bet ; bridge to GARCH
Θ Theta	passage of time	-1.8/day	the option bleeds ; buyers pay, sellers earn
ρ Rho	rate r	9.8/rate-pt	matters mainly for long-dated options

(Values for the 3-month at-the-money S&P call.)

PUT–CALL PARITY

The Greeks describe one option in isolation. One last relation ties calls and puts to *each other* — and needs no pricing model, only the absence of arbitrage:

$$C - P = S - Ke^{-rT}$$

A call minus a put (same strike and expiry) must equal the stock minus the discounted strike — otherwise a riskless profit exists.

Check on our S&P options

$$C - P = 296.8 - 222.1 = 74.7, \quad S - Ke^{-rT} = 7483 - 7483e^{-0.04 \times 0.25} = 74.5.$$

Consequence: once you can price a call, the put comes for free — and the two markets are kept consistent by arbitrage.

WHY CONTINUOUS TIME?

So far everything has been **discrete** — one observation per day. The theory behind option pricing, though, is written in **continuous time**, where the price moves over tiny instants dt .

- Two ingredients: the **Brownian motion** that drives the randomness, and the **Geometric Brownian Motion (GBM)** price model built on it.
- GBM is the engine behind **Black–Scholes** — and a clean way to see, one more time, what makes real markets harder than the textbook.

We keep it intuitive: the picture and the two assumptions, no stochastic calculus.

BROWNIAN MOTION: THE BUILDING BLOCK

The idea

Brownian motion W_t is the continuous-time limit of a random walk — pure, structureless randomness accumulating over time.

Its defining properties:

- it starts at 0, and its steps over non-overlapping periods are **independent**;
- a step over a tiny interval dt is **normal** with variance dt , so the standard deviation grows like $\sqrt{\text{time}}$ (the “square-root-of-time” rule);
- the path is **continuous but nowhere smooth** — it jitters endlessly, like a price on a screen.

GEOMETRIC BROWNIAN MOTION: THE MODEL OF PRICES

The idea

A price's *proportional* change is Brownian:

$$\frac{dP_t}{P_t} = \mu dt + \sigma dW_t,$$

with **drift** μ (expected growth) and **volatility** σ .

Two consequences, both familiar from earlier in the course:

- the **log price is a random walk with drift** (an $I(1)$ series);
- **log returns are normal with constant volatility** σ , and prices are *lognormal* — so they can never fall below zero.

WHAT GBM SAYS ABOUT PRICES AND RETURNS

Because the *proportional* change is Brownian:

- **prices are lognormal** — right-skewed and bounded below by zero (a price can fall to, but not below, 0: the model respects limited liability);
- the **mean sits above the median** — a few big up-moves drag the average above the typical outcome.

A concrete picture — the S&P one year out

Spot 7483, drift 13%, volatility 17.4%: the **mean** outcome is 8523, the **median** 8395, and the **90% range** is [6310, 11170] — a $\pm 25\%$ band. The *uncertainty dwarfs the drift*.

ESTIMATING GBM'S TWO NUMBERS

GBM has just two parameters, and both come straight from daily returns:

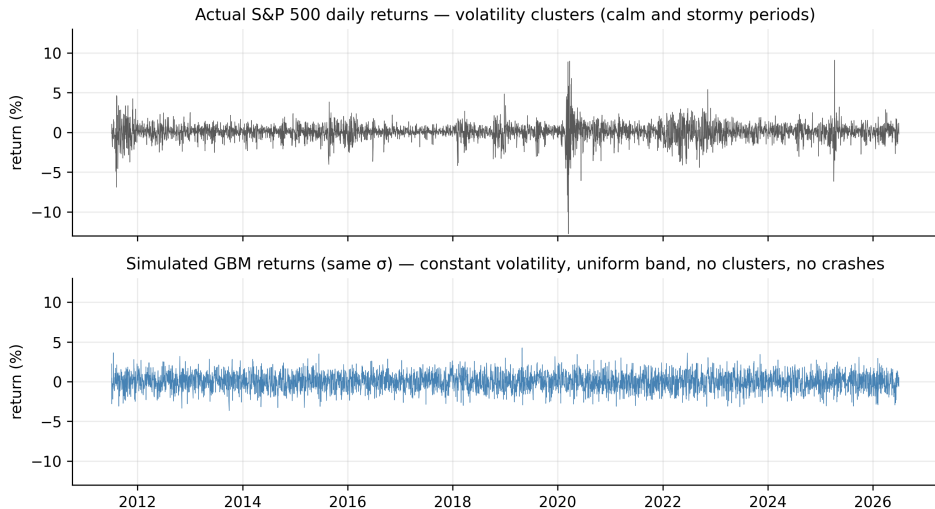
$$\hat{\sigma} = s_{\text{daily}} \sqrt{252} \quad (\text{annualised volatility}), \quad \hat{\mu} = \bar{r}_{\text{daily}} \cdot 252 + \frac{1}{2} \hat{\sigma}^2 \quad (\text{drift}).$$

Ticker	volatility $\hat{\sigma}$	drift $\hat{\mu}$
AAPL	28.4%	26.5%
SPX	17.4%	13.0%
GLD	16.8%	7.7%

The single stock is far more volatile than the index — diversification at work. Annualising uses 252 trading days; the $\frac{1}{2} \hat{\sigma}^2$ term is the drift correction from moving between prices and log prices.

WHERE GBM BREAKS (1): VOLATILITY IS NOT CONSTANT

GBM assumes one fixed σ forever. Simulate that beside the real S&P:



The simulation (bottom) is a **featureless, uniform band**; reality (top) **clusters** — calm stretches, violent ones, the 2020 crash towering over all. That clustering is exactly what **ARCH / GARCH** captures.

WHERE GBM BREAKS (2): THE TAILS ARE TOO FAT

Under GBM a big daily move is essentially impossible — yet every series has crash days that are enormous in σ units:

Ticker	worst day	size under GBM	GBM probability
AAPL	-13.8%	-7.7σ	~ 1 in 100 trillion
SPX	-12.8%	-11.7σ	~ 1 in 10^{31}
GLD	-10.8%	-10.2σ	~ 1 in 10^{24}

The S&P's worst day is an 11.7σ event whose GBM probability has thirty zeros after the decimal — it “should never occur,” yet it did. **Fat tails, one more time.**

THE CONTINUOUS-TIME FIX: ADD JUMPS

The clean way to let a continuous-time model *crash* is a **jump-diffusion**:

smooth GBM part + occasional sudden jumps.

- **Most days:** ordinary GBM diffusion.
- **Rarely:** a sudden jump — a gap down (or up) that no smooth path could produce.

Adding jumps restores the fat tails GBM is missing — the continuous-time cousin of the discrete-time tools we built earlier (**GARCH** for clustering, **EVT** for the tail).

FROM GBM TO BLACK-SCHOLES, IN ONE BREATH

Why does GBM matter for options? Because it lets you **price** them.

- Under GBM, an option can be **perfectly hedged** by holding Δ units of the stock (the delta hedge from the Greeks) and rebalancing.
- A perfectly hedged position is riskless, so by **no arbitrage** it must earn the risk-free rate — and that one condition *pins down* the option's price.
- That price is the **Black–Scholes formula**. The only input you cannot observe is σ — which is why forecasting volatility matters.

KEY TAKEAWAYS

- An **option** is the right (not obligation) to buy (**call**) or sell (**put**) at strike K by expiry T , for a premium; four building blocks — long/short call/put.
- Options vary by **type**, **exercise style (European/American)**, **moneyness**, **maturity**, **payoff**; Black–Scholes prices *European* options.
- At expiry, value = payoff $\max(S_T - K, 0)$ / $\max(K - S_T, 0)$; before expiry, price = **intrinsic** + **time value**.
- Of the six pricing inputs, **volatility** is the crucial (and only unobservable) one — higher σ raises calls *and* puts.
- The **Greeks** are the sensitivities you hedge: Δ (underlying), Γ (curvature), ν (volatility), Θ (time), ρ (rates). **Vega is the bridge to the volatility models.**
- **Put–call parity** $C - P = S - Ke^{-rT}$ ties calls and puts by no-arbitrage.
- **Continuous time:** prices follow **GBM** (normal log returns, constant σ) — the base for **Black–Scholes**; its gaps (clustering, fat tails) are what GARCH and EVT address.

Questions?